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Reversed domains in x-cut lithium niobate thin films

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ABSTRACT

Periodically poled domain inversion structures with a period of 20 μm had been prepared on x-cut lithium niobate films using the external electric field. The effects of voltage and pulse number on the polarization reversal were investigated using the piezoresponse force microscopy (PFM), and the temperature stability and depth distribution of the reversed domains were studied. The minimum electric field intensity for the domain inversion growth was 28.2 kV/mm, which was higher than that of z-cut bulk lithium niobate (21 kV/mm). When the electric field intensity was large, such as 36.9 kV/mm, the reversed domain had obvious lateral expansion. Increasing the pulse number could cause the lateral expansion and the mergence of the domains. A proper choosing of voltage, pulse number and pulse duration would result in a domain with a 50% duty cycle. The temperature stability of the reversed domain was tested. When the annealing temperature was below 300°C, there was no obvious change in the domain reversal regions. When the annealing temperature was 400°C, the shape of the micro-domain structure near the negative electrode regions changed. The cone-shaped small reversed domains near the negative electrode regions disappeared after annealed for 3 h. There was no further change of the reversed domain when the annealing time exceeded 3 h. Argon ion etching was used to gradually strip the film, and the shape of the reversed domains at different depths was measured. When the domain inversion region was 200 nm below the surface, there was almost no change compared to the domain at the surface. As the depth became deeper, the domain reversal area became smaller. Compared to domain at the surface, the domain width at a depth of 412 nm was reduced by 0.5 μm in width.

1. Introduction

Lithium niobate had a wide light transmission range, excellent electro-optical, acousto-optic, and nonlinear optical properties, and was a widely used optical crystal [1–4]. The lithium niobate on the insulator (LNOI) had high refractive index contrast, which could result in smaller device size, and higher integration density [5,6]. Therefore, LNOI was considered as the ideal platform for integrated optics [7]. Periodically polarized lithium niobate was an important way to achieve quasi-phase matching, because it could use the largest non-linear coefficient d_{33} (d_{33} = 31.5 pm/V) [8,9]. Quasi-phase matching compensated the phase mismatch caused by refractive index dispersion by periodically reversing the direction of crystal polarization, thereby achieving efficient wavelength conversion [10]. In recent years, excellent works had been done on quasi-phase matching to achieve nonlinear optics on lithium niobate films, and efficient frequency conversion had been achieved [11–22]. A high-quality domain structure played an important role in the efficient frequency conversion, and the study on domain growth would provide important information to nonlinear optics. This paper mainly explored the domain inversion of x-cut lithium niobate films using the external electric field. Periodically poled domains with a near 50% duty cycle were obtained, and the smallest polarization field strength was observed to be 28.2 kV/mm. The domains were stable at temperature up to 300°C. When the temperature was 400°C, the shape of the reversed domain changed near the negative electrode. The reversed domain shapes at different depths below the surface were explored. Within 203 nm, the domain pattern was almost unchanged compared to that at the surface. When at a depth of 412 nm, the domain reversal areas were reduced. Compared with the domain pattern at the surface, it was reduced by about 0.5 μm in the lateral direction. These results provided information for the preparation and application of periodically polarized lithium niobate thin films.

2. Experiment and result

Lithium niobate on insulators were prepared by ion implantation and

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direct bonding techniques (by NANOLN). It consisted of a 500-μm thick lithium niobate substrate, a 2-μm thick silicon dioxide insulating layer, and a 500-nm thick lithium niobate film. The preparation process was shown in Fig. 1. In order to achieve external electric field polarization, a 70-nm thick chromium film was deposited on the surface of the lithium niobate film by electron beam evaporation. The electrode with a period of 20 μm was defined by photolithography, and then chromium was wet-etched. The distance between the electrodes was 9.2 μm. The cross-section of the electrode on the x-cut LNOI was shown in Fig. 2 (a).

The polarization inversion on lithium niobate thin film was achieved with multiple electric pulses [11,23]. This process could stabilize the reversed domains, and because the pulse duration was short, the current thermal effect could be avoided. A probe station was used to apply high-voltage pulse to the electrode. The connecting circuit was shown in Fig. 2(b). The physical environmental-condition had a significant impact on domain reversal. The coercive electric field of lithium niobate would be significantly reduced at high temperatures [24]. As the humidity increased, the size of the domains increased nonlinearly, which was reported in Refs. [22]. During the experiments, all of the following polarization inversion were performed at room temperature (24°C), and the air humidity was 45% RH. In addition, in order to avoid air breakdown caused by high electric field strength, the samples were placed in silicone oil. An example of the waveform of the electric pulse was shown in Fig. 2(c). The period of the polarization pulse was selected to be 0.1 s, and the pulse duty ratio (defined as the ratio of the duration of a single pulse to the period) was 50%. The reversed domains were observed using a PFM (model: Dimension Icon, Manufacturer: Veeco Instruments Inc.). When an excitation voltage was applied to the crystal, because of the inverse piezoelectric effect, the domain inversion regions and the non-inversion regions exhibited different behaviors (shrinkage or expansion) due to different signs of inverse piezoelectric coefficients, thereby distinguishing the inversed and non-inversed domains.

(1) Effect of voltage and pulse number on polarization inversion

When the inversion voltage was 230 V, no polarization reversal was observed. After that, the pulse voltage was increased and it was found that the lowest domain inversion voltage was 260 V (28.2 kV/mm), which was larger than the bulk lithium niobate (~21 kV/mm). One possible explanation was that the Li^+ out-diffusion caused by annealing during the fabrication processes of the thin film and the bonding interface between the thin-film LN and the SiO_2 buffer layer increased the coercive field [14,25] (As shown in Fig. 3(a), the yellow regions were the reversed regions, and the black regions were the non-reversed regions.). Fig. 3(a) (260 V) showed that the domain started to reverse from the positive electrode and extended to the negative electrode, and the reversed domain was sharp and tapered, wide in the original +z direction of the crystal, and gradually narrowed in the direction extending toward -z. When the voltage rose to 280 V, the reversed domains in the same reversed region merged with each other, but it was not completely reversed near the negative electrode. When the voltage was further increased to 340 V, it could be observed that the domain inversion regions near the positive electrode had grown laterally. As the voltage continued to increase, the domain inversion regions gradually became a reversed trapezoid. The reversed domain width at the position of the middle distance between the positive and negative electrodes was measured at different poling voltage. The result was shown in Fig. 3 (b). The black dots were measured data, and the black line was the linear fit. As the pulse voltage increased, the domain's lateral extension became more obvious, and the expansion width was approximately proportional to the voltage increase.

Next, the effect of pulse number was investigated. The result was shown in Fig. 4. The pulse parameters were: voltage = 270 V, pulse period = 0.1 s, duty cycle = 50%, and the pulse number was set to 3, 10, 300, and 1000, respectively. The pulse number did increase the growth of the domains. As the pulse number increased, the adjacent cone-shaped domains gradually merged, and at the same time, they gradually grew towards the -z direction of the electrode. However, it could be found from the figure that although the pulse number was increased, there were still some non-inversion regions, and the lateral expansion of the domains was obvious near the positive electrode.

3. Temperature stability

Sometimes, non-linear optical devices worked at high temperature. Therefore, it was necessary to study the temperature stability of the reversed domain. The reversed domains were annealed in air, and then the shape of the reversed domains was observed using PFM. The result was shown in Fig. 5, the pulse parameters were: voltage = 320 V, pulse period = 0.1 s, duty cycle = 50%, and pulse number = 5. Before the annealing, the domains were completely reversed near the positive electrode. The domains were not completely reversed near the negative electrode, and it showed a series of spike-like isolated shape. When the sample was annealed at 200°C and 300°C for 3 h in air, respectively, the shape of the reversed domain did not change. When the annealing temperature reached 400°C and the annealing time was 3 h, the spike-like reversed domain near the negative electrode disappeared. The reversed domains of bulk lithium niobate could remain stable at a

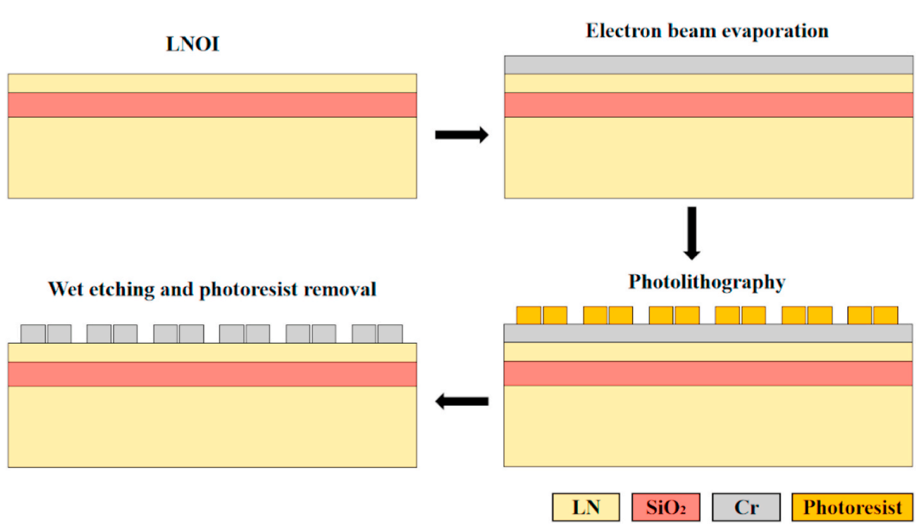

Fig. 1. Fabrication process for the electrodes.

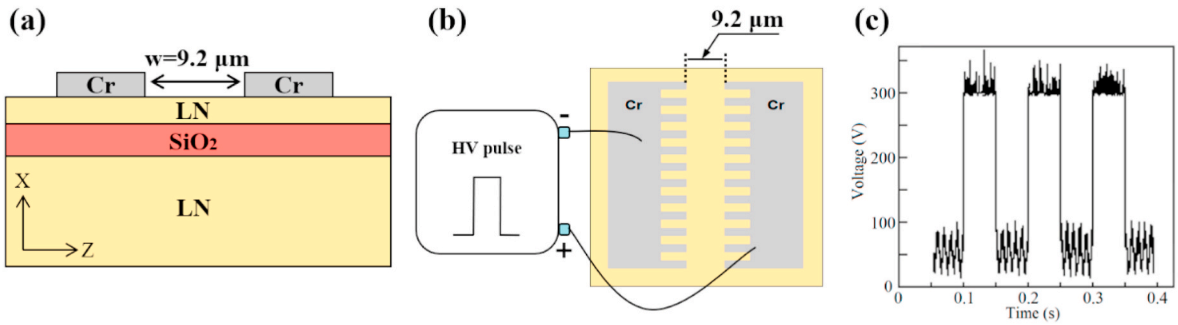

Fig. 2. (a) A cross-section of the electrode on the LNOI. (b) Schematic diagram of polarization circuit and (c) measured pulse waveform of the pulse power supply.

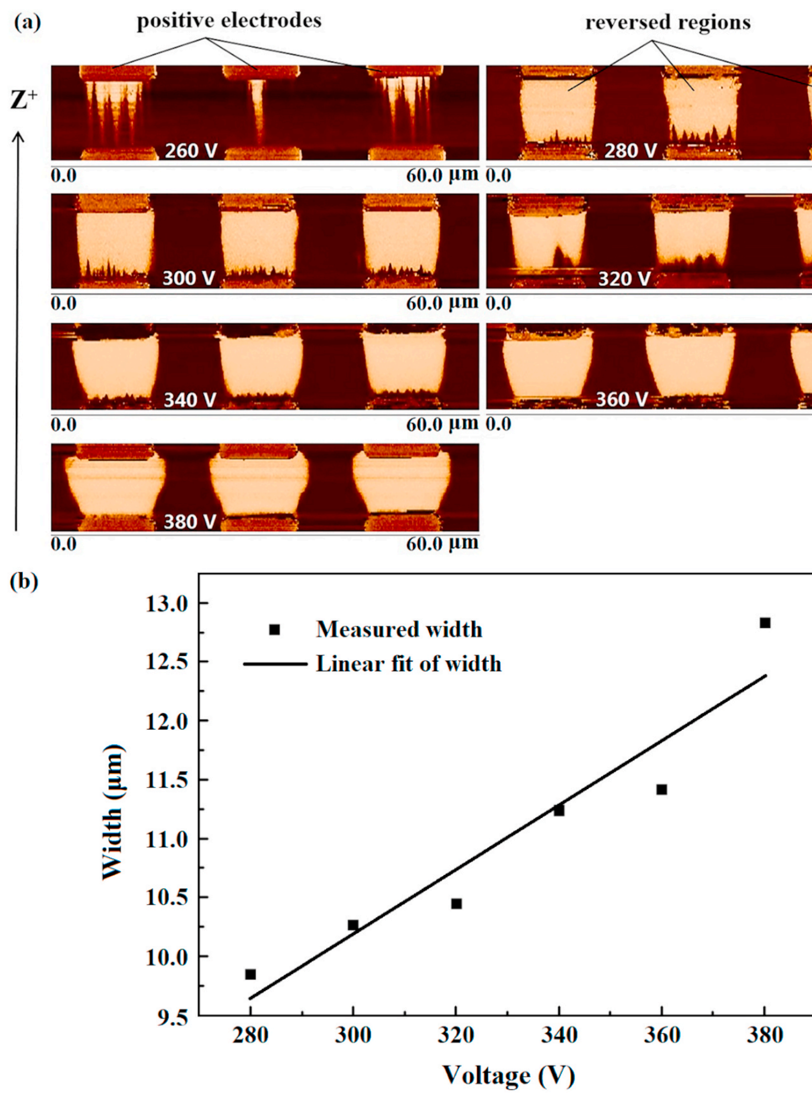

Fig. 3. (a) Domain inversion images under piezoelectric microscope when the pulse period was 0.1 s, the pulse number was 5, and the polarization voltages were 260, 280, 300, 320, 340, 360, and 380 V, respectively. (b) Measured domain inversion width in the middle of the electrode as a function of voltage and a linear fit.

temperature of 500°C [23]. One possible explanation was that the defects introduced by ion implantation in the film preparation process result in a higher internal electric field which made it easier for the reversed domain to return to the original polarization direction when the temperature was high enough. After that, the annealing time was further increased, but the shape of the reversed domain did not change. It seemed that the small reversed domain was not stable at high temperature. The sample continued to be annealed at 500°C for 3 h in air, and some reversed domains changed slightly and some reversed domains did not change.

(3) Domain inversion depth distribution

When light propagated in the waveguide, the optical mode was distributed throughout the thickness of the lithium niobate film, so the conversion efficiency of nonlinear optical process was closely related to

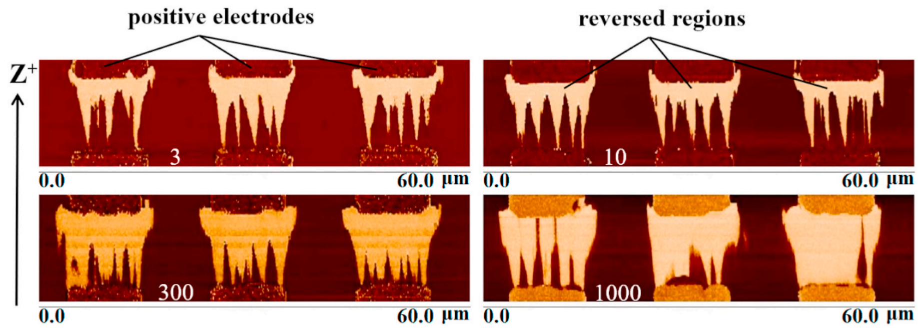

Fig. 4. Domain inversion patterns when the pulse number were 3, 10, 300, and 1000, respectively.

the depth distribution of the domains. In order to investigate the depth distribution of the reversed domain, the argon ions were used to etch lithium niobate films, and then the shape of the reversed domain at different depth was characterized by PFM. The results were shown in Fig. 6, and its polarization inversion parameters were: voltage = 280 V, pulse period = 0.1 s, duty cycle = 50%, and the pulse number = 5. From Fig. 6, the domain inversion regions at 203 nm below the surface was almost identical to that at the surface. At 360 nm below the surface, the area of the reversed domain had been slightly reduced. The domain reversal area was reduced in both the horizontal and vertical directions, and the horizontal change was more severe, and it was about 0.4 μm. Its reduction in the horizontal direction was 0.5 μm at 412 nm below the surface. The width of the reversed domain at different depth was shown in Fig. 6(b). The lateral width gradually decreased with increasing depth, and there was an approximately linear correlation between the lateral width and depth. We considered that the domain reverse process in LNOI was similar to that in bulk material. In bulk lithium niobate, the domain growth process included arising of new domains, forward growth of the domains, sideways domain wall motion, and coalescence of residual domains [26]. The forward growth of isolated needle-like domains in the polar direction was a result of step generation and kink growth [27]. The mechanism of tapered shape in depth direction might also be explained in the same way. Furthermore, the electric field reduced with depth [25], which could reduce the reversed domain size. And the trapezoidal shape or tapered shape of the reversed domain was reported in other literatures, including the bulk lithium niobate [28] and the x-cut lithium niobate thin film [29].

4. Conclusion

In conclusion, the effects of voltage and pulse number on the polarization reversal were investigated, and the temperature stability and depth distribution of the reversed domains were studied. The smallest polarization field strength was about 28.2 kV/mm, which was higher than z-cut bulk lithium niobate. When the voltage was high, such as 340 V, the reversed domain appeared to expand laterally, and the width of

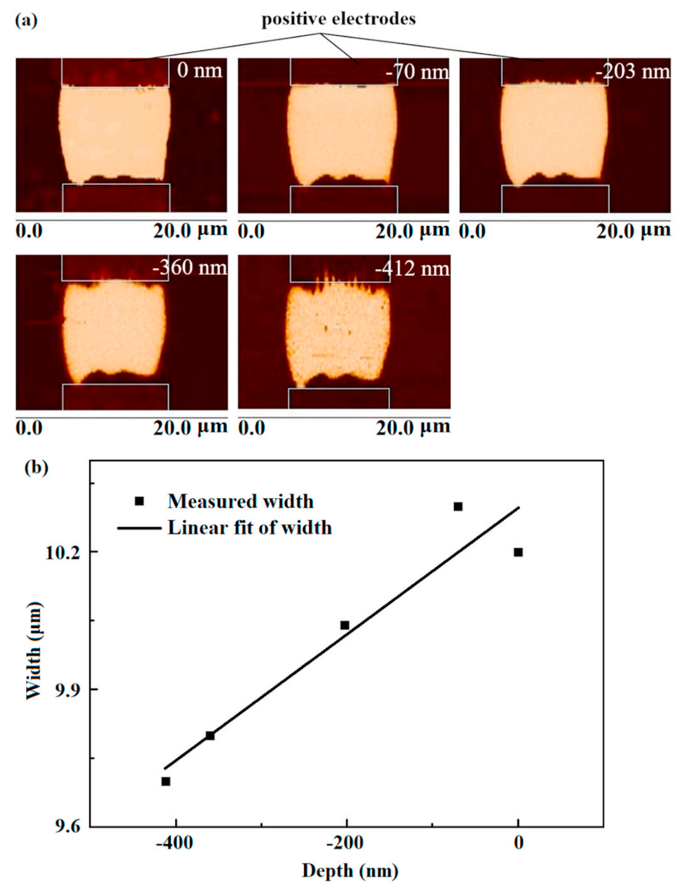

Fig. 6. (a) PFM pictures of domain inversion at different depths below the film surface. (b) Measured domain inversion width in the middle of the electrode as a function of depth and a linear fit.

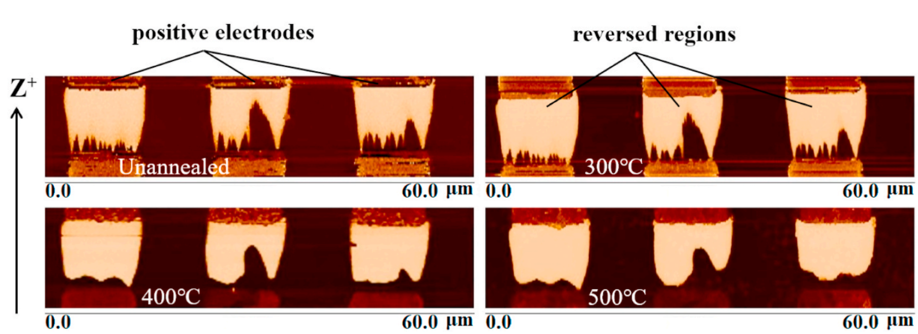

Fig. 5. Domain inversion pattern before annealing and domain inversion pattern after annealing at 300, 400, 500°C in sequence.

the lateral expansion was proportional to the voltage. The reversed domain did not change at annealing temperature of 300°C for 3 h. After annealing at 400°C for 3 h, the cone-shaped small domain structure at the negative electrode region disappeared, and there was no further change if the annealing time exceeded 3 h. Finally, the domain inversion patterns at different depths were explored, and the lateral width of domain inversion gradually narrowed with increasing depth. These results provided useful information for the preparation and the study of periodically poled domain inversion structures.

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CRediT authorship contribution statement

Honghu Zhang: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Houbin Zhu:** Resources, Writing - review & editing. **Qingyun Li:** Writing - review & editing. **Hui Hu:** Conceptualization, Formal analysis, Writing - review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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